

$\because K$ is an integer, $7K+2$ is an integer.

By Defn 1, $7n+4$ is even. $\therefore q$ is True. By defn. of conditional, $(p \rightarrow q)$ is True.

Direction 2: We shall prove $(q \rightarrow p)$ is True by contradiction.

$\therefore$ We shall prove $(q \wedge \neg p) \rightarrow F$ is True.

Let $(q \wedge \neg p)$ be True. By defn. of AND, q is True and $\neg p$ is True

$\neg p$: n is not even, i.e. n is odd (Defn. 1)

$\therefore (7n+4)$ is even, n is odd.

By Defn 1, $\exists$ integers K_1 and K_2 such that $(7n+4) = 2K_1$ $n = 2K_2+1$

$\therefore (7n+4) - n = 2K_1 - (2K_2+1) = 2(K_1 - K_2 - 1) + 1$. It's easy to see that $(K_1 - K_2 - 1)$ is an integer. $\therefore (7n+4) - n$ is odd, from Defn 1.

Also, $(7n+4) - n = 6n+4 = 2(3n+2)$. It's easy to see that $(3n+2)$ is an integer.

By Defn 1, $(7n+4) - n$ is even.

From Defn 1, an integer can't be both even and odd. $\therefore$ This is a contradiction.

$\therefore \neg p$ is False. By defn. of AND, $(q \wedge \neg p)$ is False. By defn. of conditional, $(q \wedge \neg p) \rightarrow F$ is True. $\therefore (q \rightarrow p)$ is True.

$\therefore (p \rightarrow q)$ is True. $\therefore (p \rightarrow q)$ is True. So, n is even if and only if $7n+4$ is even $\square$.

2g) Prove that if n is a positive integer, then n is odd if and only if $5n+6$ is odd.

Soln: p : n is an odd integer q : $5n+6$ is an odd integer

We shall prove $p \leftrightarrow q$ is True. $\therefore$ We need to show $(p \rightarrow q) \wedge (q \rightarrow p)$ is True.

$\because p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$. By the defn. of AND, $(p \rightarrow q)$, $(q \rightarrow p)$ are both True, we need to show that.